

Himasha Minoli Valantina

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👤 PROFESSIONAL SUMMARY

Dedicated BSc(Hons) Computer Science undergraduate at SLIIT with a focus on AI-powered systems and compiler design. Proven experience in full-stack development and integrating Large Language Models (LLMs) like Gemini into scalable applications. Eager to apply theoretical knowledge in Operating Systems and automation to contribute to innovative open-source ecosystems.

🎓 EDUCATION

BSc (Hons) Computer Science *Undergraduate*
Sri Lanka Institute of Information Technology (SLIIT) 2023 Nov – Present

Key Modules: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Distributed Systems, Statistics, Computer Networks, Programming Paradigms, Engineering Mathematics, Human Computer Interactions.

G.C.E. Advanced Level (Physical Science Stream): 1 B Pass, 1 C Pass, 1 S Pass **2022**
Holy Cross College, Gampaha

G.C.E. Ordinary Level: 4 A Passes, 3 B Passes, 2 C Passes **2019**
Holy Cross College, Gampaha

🔧 TECHNICAL PROJECTS

EduGemini (AI-Powered Learning Platform) — *Gemini API, React, Node.js, Tailwind CSS* Jan 2026
[Contribution: AI Integration, Full Stack Development]

- **Full-Stack AI Orchestration:** Architected an end-to-end educational platform using the Gemini 1.5 Flash API and Node.js to transform user prompts into structured, curriculum-aligned study modules with a high-performance React frontend.
- **Prompt Engineering Reliability:** Engineered a robust prompt-layer to enforce deterministic outputs, successfully mitigating model hallucinations and optimizing the generation of high-accuracy pedagogical content.

Health Vault — *Ballerina, React, TypeScript, Vite* Aug 2025
[Contribution: Back-end]

- Engineered a network-aware backend using Ballerina to power a digital "Child Health Development Record" system. Built a type-safe, high-performance React/TypeScript interface to track vaccination schedules and growth metrics.

Serango (AdaptIQ Hackathon Finalist) — *Java Spring Boot, MongoDB(Vibe code)* Oct 2025
[Contribution: Back-end]

- Selected as a Top 10 Finalist (out of 40+ teams) in a 22-hour onsite coding challenge to solve tourism industry problems. Rapidly prototyped a functional MVP that successfully adapted to 8 dynamic constraints introduced during the competition.

TestLang Compiler (Custom DSL for Automation) — *C++, Lex, Yacc, Compiler Design* Dec 2025
[Contribution: Language Design, Compiler Engineering]

- **Compiler Development:** Engineered a custom Domain Specific Language (DSL) from scratch using C++ and Lex/Yacc to streamline automated test script generation and execution.
- **Syntax Semantic Analysis:** Architected the complete compiler pipeline, including lexical analysis, parsing, and an Abstract Syntax Tree (AST) to transform high-level test logic into executable machine-ready instructions.

🏆 ACHIEVEMENTS

- **AdaptIQ Hackathon (IIT)** — *Top 10 Finalist* 2025
Selected to compete in the final 22-hour physical development challenge organized by WIEEE. (Team: Team Nybble).
- **Innovate with Ballerina 2025** — *Winner (Social Media Contest)* 2025
Organized by IEEE University of Moratuwa & WSO2. (Team: Ballerina Cappuccina).
- **Code Quest v1.0** — **IEEE Sabaragamuwa University** — *Finalist (Top 10)* 2025
– Successfully advanced to the final round as one of the Top 10 teams by demonstrating strong problem-solving and algorithmic thinking (Team: ObscurusSquad)

CERTIFICATIONS

- **Certificate in AI/ML Engineer - 1** — *SLIIT* 2025
- **Certificate in AI/ML Engineer - 2** — *SLIIT* 2025

VOLUNTEERING & LEADERSHIP

- **SLIIT Final Exams Support Series** — *Peer Tutor* 2024
 - Volunteered to conduct academic revision sessions to assist peers in preparing for final examinations.

REFERENCES

Mr. Dinesh kumara
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Mr. Lakindu Perera
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